Eric Alejandro Rozan, Master in Physics

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Education

2022-Ongoing — **PhD Degree in Physics** – Instituto Balseiro, CNEA - UNCuyo, Argentina.
• Thesis title: *Mathematical approaches to epidemiological outbreaks: analysis of data and models*.
Supervisors: Dr. Marcelo Kuperman and Dr. Sebastian Bouzat.

2021-2022 — **Master Degree in Physics** – Instituto Balseiro, CNEA - UNCuyo, Argentina.
• Thesis title: *Formulation of Advanced Epidemiology Models*.
Supervisor: Dr. Marcelo Kuperman.
• Obtained Full Scholarship from National Atomic Energy Commission (CNEA).

2012-2020 — **Bachelor Degree in Physics** – Facultad de Ciencias Exactas y Naturales, Universidad de Buenos Aires, Argentina.
• Thesis title: *Emergency Evacuation in the Presence of Social Groups*.
Supervisors: Dr. Claudio Dorso and Dr. Guillermo Frank.
Thesis worthy of the *Premio Másperi 2021* award given by Argentine Physics Association (AFA).

2006-2011 — **Bilingual Secondary School Graduate** – Escuela Secundaria Integral Scholem Aleijem.
• Awarded Diploma of "Best Student of the Cohort in the Economy and Organization Management Orientation".

Publications

Article under review — **Analysis of non pharmaceutical interventions with SIR epidemic models: decreasing the infection peak vs. minimizing the epidemic size**, *E.A. Rozan, M.N. Kuperman & S. Bouzat*, preprint available at doi 10.48550/arXiv.2604.08420.

Article under review — **Modeling Epidemics on Multiplex Networks: Epidemic Threshold and Basic Reproduction Number**, *E.A. Rozan, M.I. Simoy, S. Bouzat & M.N. Kuperman*, preprint available at doi 10.48550/arXiv.2510.12614.

Journal Article – 2026 — **Exact and Approximate Constants of Motion in Stochastic Contact Processes**, *D.H Zanette & E.A. Rozan*, Entropy 2026, 28(3), 276, doi 10.3390/e28030276.

Journal Article – 2024 — **The importance of the incubation time distribution in compartmental epidemiological models**, *E.A. Rozan, M.N. Kuperman & S. Bouzat*, Physica A: Statistical Mechanics and its Applications, Volume 650, 129998, doi 10.1016/j.physa.2024.129998.

Journal Article – 2023 — **Testing Lockdown Measures in Epidemic Outbreaks Through Mean-Field Models Considering the Social Structure**, *E.A. Rozan, S. Bouzat & M.N. Kuperman*, Physica A: Statistical Mechanics and its Applications, Volume 632, Part 1, 129330 doi 10.1016/j.physa.2023.129330.

Publications (continued)

Journal Article – 2023 — **Emergency Evacuations of Social Groups**, *E.A. Rozan, G.A. Frank, F.E. Cornes, I.M. Sticco & C.O. Dorso*, Anales de la Asociación Física Argentina, Vol. 33, Nro. 4, doi 10.31527/analesafa.2022.33.4.112 .

Journal Article – 2022 — **Microscopic dynamics of escaping groups through an exit and a corridor**, *E.A. Rozan, G.A. Frank, F.E. Cornes, I.M. Sticco & C.O. Dorso*, Physica A: Statistical Mechanics and its Applications, Volume 597, 127271, doi 10.1016/j.physa.2022.127271.

Scientific Event Organization

May 2026 — **XXIII Regional Workshop of Statistical Physics and Applications to Condensed Matter, "TREFEMAC 2026"** – *Simons Foundation & Instituto Balseiro*.
Member of the Organizing Committee of the international workshop, and the associated post-graduate course *"Active Matter and Dissipative Structures"*.

Conference Presentations

Oct 2026 — **Workshop on Interdisciplinary Complex Systems** – *Physics Department of the University of Buenos Aires & University of San Andrés.*

- Poster Presentation *"Modeling epidemic outbreaks in multiplex networks: epidemic threshold and basic reproduction number"* by E.A. Rozan, M.I. Simoy, S. Bouzat & M.N. Kuperman.
- Poster Presentation *"Influential nodes: a selection of indices"* by C.A. Huaylla & E.A. Rozan.

May 2026 — **XXIII Regional Workshop of Statistical Physics and Applications to Condensed Matter, "TREFEMAC 2026"** – *Simons Foundation & Instituto Balseiro*.

- Poster Presentation *"Analysis of interventions for mitigating epidemic outbreaks: decreasing the infection peak vs. minimizing the epidemic size"* by E.A. Rozan, M.N. Kuperman & S. Bouzat.
- Poster Presentation *"Analyzing various indices to identify influential nodes in complex networks"* by C.A. Huaylla & E.A. Rozan.
- Poster Presentation *"Axelrod's Swarmalators"* by E.A. Rozan.

May 2025 — **XXII Regional Workshop of Statistical Physics and Applications to Condensed Matter, "TREFEMAC 2025"** – *Universidad Nacional de Salta*.
Oral Presentation *"Incubation time distributions in epidemiological models and their impact on key indicators"*.

Dec 2024 — **Dynamic Days Latin America and The Caribbean 2024** – *Universidad de Buenos Aires.*
Poster presentation *"Impact of Incubation Time Distributions on Epidemic Model Predictions"* by E.A. Rozan, S. Bouzat & M.N. Kuperman.

Oct 2024 — **XXVIII Research Conference of the National University of Cuyo** – *Universidad Nacional de Cuyo.*
Oral presentation *"Mathematical approaches to epidemiological outbreaks: analysis of data and models"*.

May 2024 — **Workshop on Dynamical Processes on Complex Networks** – *ICTP-SAIFR.*
Poster Presentation *"Testing Lockdown Measures in Epidemic Outbreaks Through Mean-Field Models Considering the Social Structure"* by E.A. Rozan, S. Bouzat & M.N. Kuperman.

Conference Presentations (continued)

May 2023 — **XX Regional Workshop of Statistical Physics and Applications to Condensed Matter, "TREFEMAC 2023"** – *Universidad Tecnológica Nacional - Los Reyunos*.
Oral Presentation *"Cyclic Isolation and Rest Periods in Mean-Field Epidemiology Models of Heterogeneous Populations"*.

Sep 2022 — **107th Meeting of the AFA** – *Asociación Física Argentina*.
Extended Oral Presentation *"Social Isolation in Mean-Field Epidemiology Models of Heterogeneous Populations"*.

Oct 2021 — **106th Meeting of the AFA** – *Asociación Física Argentina*.

- Extended Oral Presentation *"Pedestrian Dynamics With Social Groups"*.
- Earned the **Premio Másperi 2021** award for the best work associated to a Bachelor Thesis defended between July, 2020 and June, 2021.

Sep 2020 — **105th Meeting of the AFA** – *Asociación Física Argentina*.
Extended Oral Presentation *"Emergency Evacuations of Social Groups"*.

Post-graduate Courses and Workshops

May 2026 — **Active Matter and dissipative structures** – *Simons Foundation & Instituto Balseiro*.
The school served as an integrated introduction to the physics of non-equilibrium systems such as active and dissipative systems, where local interactions and the continuous flow of energy give rise to emergent behaviors that have no equilibrium analogue.

May 2025 — **II School on Modelling Infectious Disease Dynamics – *ICTP-SAIFR***.
Participation as monitor and assistant in the continuation to the School organized the by the *International Centre for Theoretical Physics - South American Institute for Fundamental Research* in São Paulo, Brazil.

May 2024 — **School on Modelling Infectious Disease Dynamics** – *ICTP-SAIFR*.
The school aimed to enhance modeling capacity in South America by teaching epidemic modeling, numerical methods, and data analysis through week-long group projects. Organized by the *International Centre for Theoretical Physics - South American Institute for Fundamental Research* in São Paulo, Brazil.

Nov 2023 — **School on Mathematical Modeling and Governance** – *ICTP-SAIFR*.
School focused on governmental management and decision-making in situations of risk and conflicts of interest, exploring the role of science and mathematical models in this process. Organized by the *International Centre for Theoretical Physics - South American Institute for Fundamental Research* in São Paulo, Brazil.

Jul 2023 — **São Paulo School of Advanced Science on Epidemic Preparedness** – *Universidade de São Paulo*.
Multidisciplinary school for graduate students specializing in the fields of exact, natural, or social sciences from different parts of the world, aiming to work on the prevention of new epidemics through interdisciplinary dialogue between various disciplines. Organized by the *University of São Paulo*.

2021 — **Statistical Mechanics: Algorithms and Computations** – *École Normale Supérieure*.
Course taken virtually through the coursera.org platform.

Science Popularization and Communication Experience

Jul.-Nov. 2016 — **Science Popularizer at Tecnópolis** – *FCEyN, UBA – Ministerio de Ciencia, Tecnología e Innovación.*
In charge of running several experiments and interactive attractions, alongside explaining the science behind them, at the Science Fair *"Tecnópolis 2016: Un Punto de Encuentro" (A Meeting Point)*, targeted for audiences of all ages. Expositor at the Math and Physics stand, and the *"Space Access"* stand of the National Space Activities Commission (CONAE)-Argentina.

2016 — **Science Popularizer at the Week Of Physics** – *Departamento de Física – Facultad de Ciencias Exactas y Naturales - UBA.*
Expositor at the *"Station of Synchronization"* of the Physics Department of the Faculty of Exact and Natural Science - University of Buenos Aires, in which High School students come with their teachers to the Faculty. In charge of explaining the physics behind several phenomena based on the synchronization concept.

Jul.-Nov. 2015 — **Science Popularizer at Tecnópolis** – *FCEyN, UBA – Ministerio de Ciencia, Tecnología e Innovación.*
In charge of running several experiments and interactive attractions, alongside explaining the science behind them, at the Science Fair *"Tecnópolis 2015: Futuro Para Siempre" (Future For Ever)* targeted for audiences of all ages. Expositor at the Math and Physics stand.

2015 — **Science Popularizer at the Week Of Physics** – *Departamento de Física – Facultad de Ciencias Exactas y Naturales - UBA.*
Expositor at the *"Station of Synchronization"* of the Physics Department of the Faculty of Exact and Natural Science - University of Buenos Aires, in which High School students come with their teachers to the Faculty. In charge of explaining the physics behind several phenomena based on the synchronization concept.

Teaching Experience

2023-Ongoing — **Graduate Teaching Assistant** – *Instituto Balseiro, UNCUYO - CNEA.*
Teaching assistant at undergraduate course of Mathematics.

High School Teacher – *Centro Educativo Toratenu.*
Mathematics Teacher at Secondary School.

Skills

Languages — Spanish - Native Language.
English - Advanced level.

Coding — Python, Matlab, LaTeX.

Misc. — Adobe Illustrator, Macromedia Flash.

Other Experiences

Nov. 2015 - Oct. 2016 — **Internship** – *Biblioteca Central Luis Federico Leloir – FCEyN, UBA.*
Participant in the proyect "**Qualitative and Quantitative Improvement of the Digital Library of the Faculty of Exact and Natural Sciences of the University of Buenos Aires (FCEyN, UBA)**", at the facility of the Central Library Luis Federico Leloir of the Faculty. In charge of digitizing, marking and ordering scientific documents.

Other Experiences (continued)

2017 - 2018 **Internship** – *Instituto de Ingeniería Bioquímica – FIUBA*.
Intern at the Faculty of Engineering of the University of Buenos Aires, in the proyect "**Optical Setup for the detection of fluorescence in microfluids**".
Supervisors: Dr. Maximiliano Perez and Dr. Oscar Martinez.
Design and building of the device for the detection of fluorescent microdrops with high-precission photodetectors.